MTH252

Timed Online Assignment – January Semester 2022

Data Structures and Algorithms II

Monday, 23 May 2022 **4:00 pm – 6:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This Timed Online Assignment (TOA) contains **SEVEN (7)** questions and comprises **SEVEN (7)** pages (including cover page).

2. You must answer **ALL** questions.

3. If you have any queries about a question, or believe there is an error in the question, briefly explain your understanding and assumptions about that question before attempting it.

4. You MUST **submit your answers via Canvas (similar to TMA submission) at the end time of this TOA** (as stated on this cover page). The 15 minutes grace period as shown on Canvas is strictly meant for technical issues encountered during submission. Thereafter, you will not be able to submit your answers and you will be considered as having withdrawn from the course. **No appeal will be allowed.**

5. You are to include the following particulars in your submission: TOA Course Code, your Full Name, and SUSS PI Number. Name your submission file as TOACourseCode_FullName_StudentPI. Use underscore and NOT Space. *Example: TOAXYZ_RaphaelLee_T1923161 (omit D/O, S/O).*

6. Your submission should consist of only one file and **must not exceed 500MB** in size. The file must be a PDF or Microsoft Word file saved in .doc or .docx format. All answers are to be typed. Flowcharts and graphs may be scanned or photographed and embedded in the PDF or Word file provided it does not exceed the file size limit of 500MB.

7. For answers which cannot be typed and require to be hand-written, ensure that question number is clearly stated on each page. All uploaded hand-written answers must be clear, readable and complete. Marks will not be awarded for un-readable or incomplete images.

8. To prevent plagiarism and collusion, your submission will be reviewed by Turnitin. The Turnitin report will only be made available to the marker and you will not be able to view it.

9. The University takes plagiarism and collusion seriously, and your Turnitin report will be examined thoroughly as part of the marking process.

10. Marks will only be awarded if FULL working is shown.

Answer all questions. (Total 100 marks)

Question 1

(a) For priority queue myP, identify the missing return value and myP contents for each operation in Table Q1(a).

Operations	**Return Value**	**myP Contents**
myP.is_empty()	True	{}
myP.add(45, A)		
myP.add(69, B)		
myP.add(80, C)		
myP.add(100, D)		
myP.add(35, E)		
myP.min()		
size(myP)		
myP.remove_min()		
myP.is_empty()		
myP.add(78, F)		
size(myP)		
myP.min()		

Table Q1(a)

(5 marks)

(b) If the priority queue myP is implemented with **heap** data structure, explain the heap tree structure construction for each "add" and "remove_min" operations in Question 1(a) step-by-step. You only need to put the key into each node of the tree.

(6 marks)

(c) Present the final heap tree structure in Question 1(b) as an array starting with index 0.

(3 marks)

(d) Analyse the time complexity of the priority queue with **heap** data structure to insert a new element if the total number is ***n***.

(1 mark)

Question 2

(a) In the map operations provided in Table Q2(a), identify the missing results and myM contents.

Operation	**Return Results**	**myM Contents**
size(myM)	0	{ }
myM[5] = 'A'		
myM[10] = 'B'		
myM[15] = 'C'		
myM[20] = 'D'		
myM[25] = 'E'		
myM[25]		
myM[15]		
myM.get(10)		
myM.get(8, 10)		
myM.get(5, 6)		
myM[30] = 'F'		
size(myM)		
del myM[20]		
myM.popitem()		
myM.pop(10)		
myM.keys()		
myM.values()		
myM.items()		
myM.setdefault(40, 'G')		

<u>Table Q2(a)</u>

(5 marks)

(b) For the map with the keys of 4, 10, 13, 18, 20, 30, develop a 10-entry hash table that results using the hash function, $\boldsymbol{h}(\mathbf{i}) = (\mathbf{i} + \mathbf{5})\ \mathbf{mod}\ \mathbf{10}$. Assume that the collisions are handled by linear probing.

(3 marks)

(c) Compute the total number of probes (key comparisons) encountered during the insertion of the 6 key values.

(2 marks)

(d) After inserting the 6 keys in Question 2(b), suppose we want to search the key value 30, compute the number of probes needed.

(2 marks)

(e) Based on the key insertions and search, explain if the current hash function is good for the given 6 keys?

(3 marks)

Question 3

(a) Insert an item with key 68 into the binary search tree shown in Figure Q3(a). Describe the binary search tree after the insertion.

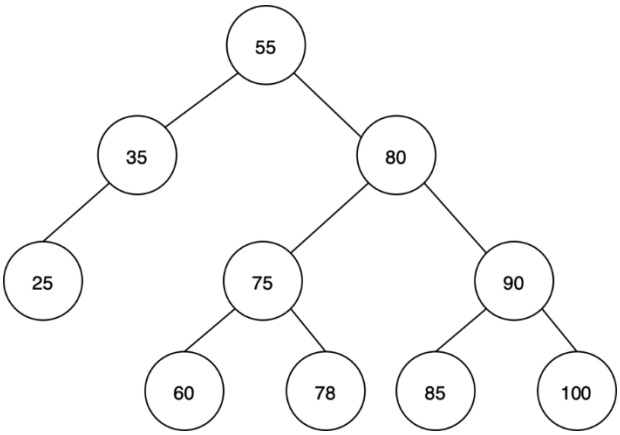

Figure Q3(a)

(5 marks)

(b) Describe AVL tree and its advantages over binary search tree. Give an example that AVL tree is more efficient than binary search tree with elements {1, 2, 3, 4, 5, 6, 7} at searching key 7.

(5 marks)

(c) After inserting key 68, is the binary search tree an AVL tree? Explain your answer.

(5 marks)

Question 4

(a) Describe divide-and-conquer strategy and list two algorithms using divide-and-conquer strategy.

(2 marks)

(b) Using the divide-and-conquer strategy, develop an algorithm (in pseudocode) to check if a given array of n integers is sorted in increasing order. Your code uses the following definition "*def checkSorted(array)*". The algorithm should recursively split the array into two equal parts and recursively check the individual parts.

(5 marks)

(c) Use Merge-Sort algorithm to sort a sequence with 8 elements 56, 55, 90, 35, 125, 15, 79, 94. Present the algorithm by using Merge-Sort Tree to show the input sequences and output sequences at nodes of the Merge-Sort Tree.

(8 marks)

Question 5

(a) The text string S is "ABABABCABABABCABD". The pattern string P is "ABABCD". Present the brute-force Algorithm to check if the pattern string P is a substring of the text string S. Calculate number of character comparisons performed.

(5 marks)

(b) The text string S is "ABABABCABABABCABD". The pattern string P is "ABABCD". Present the Knuth-Morris-Pratt Algorithm to check if the pattern string P is a substring of the text string S. Calculate number of character comparisons performed.

(9 marks)

(c) Compare the algorithms in Questions 5(a) and 5(b), analyse which one is more efficient to search P in S?

(1 mark)

Question 6

In the graph in Figure Q6, the single source is the vertex A. Present the steps of Dijkstra's Algorithm to obtain the shortest path tree from vertex A to vertex G. What is the total weight of the shortest path tree?

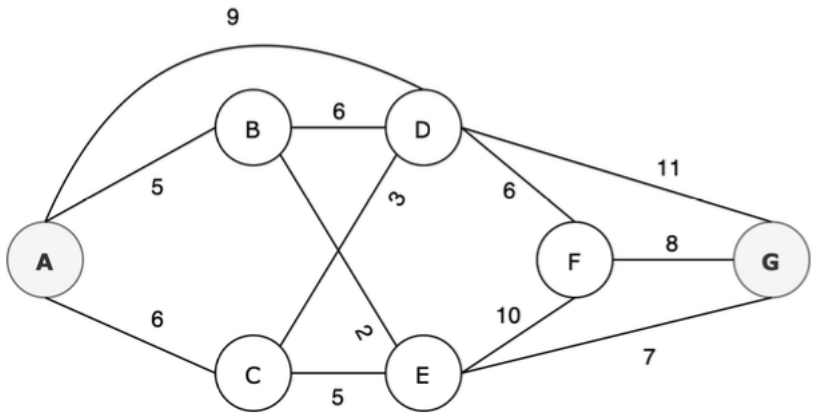

Figure Q6

(15 marks)

Question 7

(a) The ABC Software builds an IT system for a local hospital. Describe the appropriate data structures for the follow subsystems and explain your answer.

(i) Customer Queue System

(ii) Patient Record System to store the historical visit records, which can be searched by NRIC.

(iii) An educational system for public to show the knowledge about different diseases (including a lot of semantic causal relationships).

(iv) An expert system for doctor to search the search the disease with given symptoms.

(4 marks)

(b) Lester pays the hospital bill of S$59, he has prepared a number of notes in the denominations of S$50, S$10, S$5, S$2. Solve the problem by providing the solution using the minimum number of notes.
(Hint: Use the dynamic programming method to find the combination.)

(6 marks)

----- END OF PAPER -----